

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 - Hackathon

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 - Hackathon	Assessment:	Assessment Tool 3 Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	Kondeti . Lahari	Reg. No.:	192325118
Section / Batch:	B. Tech AIML	Date:	31-08-2026

Code of Conduct

I K. Lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Lahari

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Employee Prediction :

Employee Attrition App

Navigation

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- Model Performance
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Using employee_attrition.csv dataset.

Fill in employee details to predict attrition:

Satisfaction Score

0.001.00

0.58

Department

Engineering

Salary

Medium

Tenure (Years)

3

Workload

Medium

Monthly Hours

180

Overtime

No

Number of Projects

4

Predict Attrition

✔ Prediction: Employee is likely to Stay

Stay Probability: 100.0% | Leave Probability: 0.0%

Fill in employee details to predict attrition:

Satisfaction Score

0.001.00

0.30

Department

Engineering

Salary

Medium

Tenure (Years)

3

Workload

Medium

Monthly Hours

180

Overtime

No

Number of Projects

4

Predict Attrition

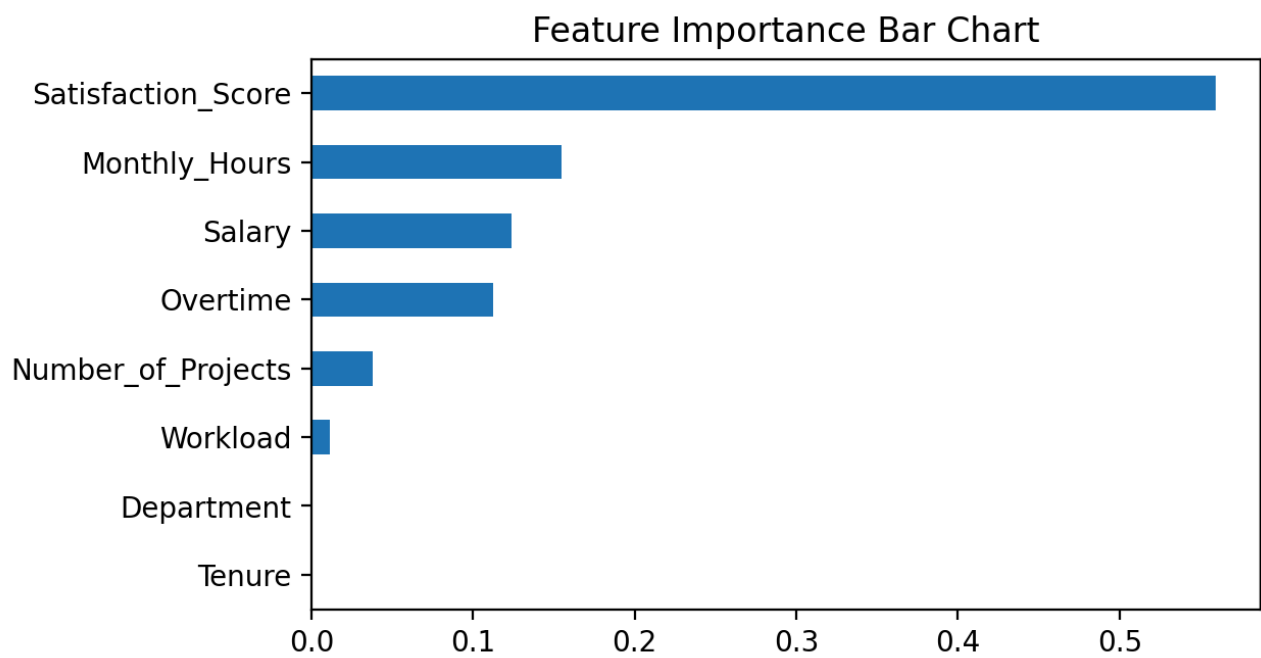
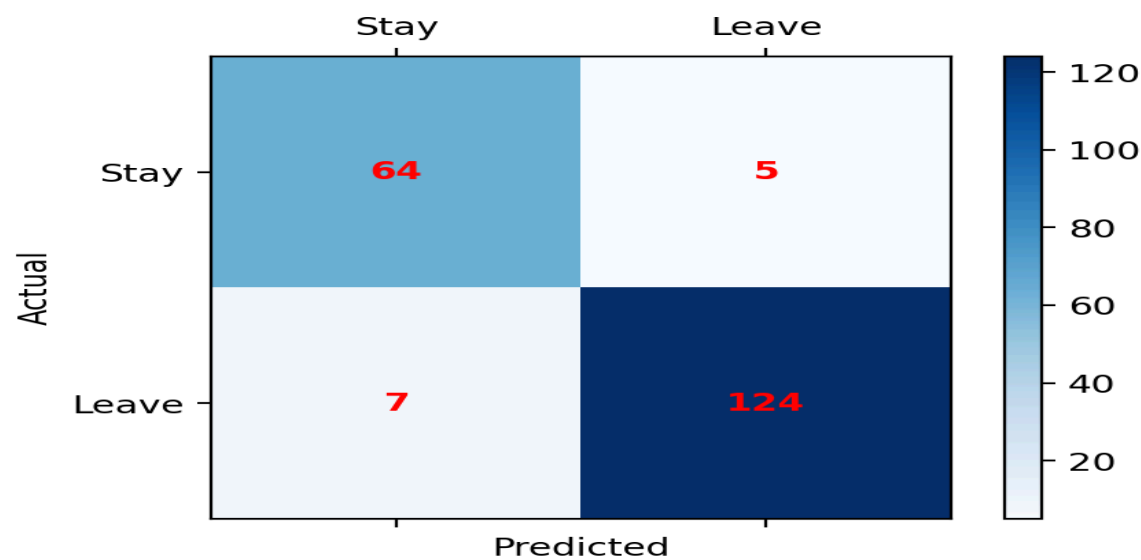
⚠ Prediction: Employee is likely to Leave

Stay Probability: 0.0% | Leave Probability: 100.0%

Reasoning / Interpretation

- Low satisfaction score (< 0.45) significantly increases attrition risk.

Graphs :



Code :

```
employee_attrition_project > app.py > ...
1 import os
2 import numpy as np
3 import pandas as pd
4 import streamlit as st
5 import matplotlib.pyplot as plt
6
7 from sklearn.tree import DecisionTreeClassifier, plot_tree
8 from sklearn.model_selection import train_test_split
9 from sklearn.metrics import accuracy_score, confusion_matrix, classifica
10
11 # Streamlit Page Setup
12 st.set_page_config(page_title="Employee Attrition Prediction", page_icon
13
14 # 1. DATASET LOADING / GENERATION
15 DATA_FILE = "employee_attrition.csv"
16
17 @st.cache_data
18 def get_data():
19     if os.path.exists(DATA_FILE):
20         df = pd.read_csv(DATA_FILE)
21         if "Tenure_Years" in df.columns:
22             df = df.rename(columns={"Tenure_Years": "Tenure"})
23         return df, False
24
25     np.random.seed(42)
```

Geo Tag Photo:

